Michael Medford, Ph.D

Data Infrastructure &
Production AI Consultant

Hands-on founding engineer and engineering leader with staff-level experience in data infrastructure and production AI. I help teams improve pipeline reliability, database performance, and cloud efficiency, and build dependable retrieval and AI orchestration systems. Available for remote consulting up to 10 hours per week.

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Skills

AI & Workflow: Agent Orchestration, Harness Engineering, Evaluation Design, Temporal, Airflow, Prefect
Architecture: Data Pipelines, Data & Event-Driven Architecture, Distributed Systems, System Reliability
Data & Backend: Python, Go, SQL, Ruby on Rails, PostgreSQL, OpenSearch, Data Modeling
Cloud & Infrastructure: AWS, GCP, Docker, Kubernetes, Terraform, Kafka

Work Experience

Founding Engineer & VP of Engineering
DillyAI · June 2026 - Present

- Built an event-sourced orchestration engine with Temporal for production LLM workflows, incorporating auditable verification pipelines
- Built AI-assisted software development workflows with parallel execution, document concurrency controls, and automated cloud execution under human-directed architecture

Staff Software Engineer - Data Systems
KoBold Metals · April 2025 - May 2026

- Built the Korecam360 data infrastructure for large-scale image ingestion, synchronization, and retrieval, supporting machine learning, historical backfills, and visualization.
- Built an operational reporting application that reduced drill superintendent reporting from multiple hours each morning to approximately 15 minutes.
- Led adoption of AI-assisted development practices, redesigned engineering workflows, and mentored engineers in AI-native development using Cursor and agent-based tools.

Staff Software Engineer - Subject Matter Expert in Datastores
Insight M · July 2024 - March 2025

- Developed a data architecture and security roadmap covering development environments, analytics pipelines, and data retention, with projected annual savings of six figures
- Optimized database processes by refactoring high-cost computational components, reducing runtime expenses, and enhancing system resiliency with comprehensive test coverage
- Enhanced data infrastructure by implementing a database event alert pipeline, launching an internal application for query plan analysis to improve query efficiency, and introducing the company's first postmortem template to foster learning from software retrospectives

Staff Software Engineer
Aumni, a JPMorgan Chase Company · November 2022 - June 2024
Promoted from Senior Engineer, July 2023

- Designed and deployed an OpenSearch indexing service for millions of documents in six weeks
- Reduced developer data synchronization time from 15 minutes to 20 seconds through infrastructure and tooling improvements
- Redesigned backend architecture with management, product, and engineering to enable flow in stream-aligned teams
- Developed a migration plan to maintain 100% uptime while refactoring a Ruby on Rails monolith into independent services within the JPMorgan ecosystem

Technical Lead of Planet Fusion Monitoring Pod
Planet Labs • March 2022 - October 2022

- Increased daily Python processing capacity 10x while reducing per-unit costs 80% in four months
- Reduced database query latency and CPU utilization 98% through SQL schema and query optimization
- Reduced cloud data-delivery processing overhead 30x using batched GCP Pub/Sub messages

Geospatial Software Engineer
Planet Labs • June 2021 - October 2022

- Implemented automated data backups & disaster recovery via terraform to comply with ISO certification
- Set up on-call notifications by integrating Sentry, PagerDuty and Slack into our Python & Go stack
- Created a comprehensive cost estimate method to measure profit margin goals for our product teams

Research Experience

A New Method for Detecting Solar System Objects on High Performance Computers
Lawrence Berkeley National Laboratory • January 2016 - September 2019 w/ Dr. Peter Nugent

- Invented planet detection pipeline that searched 100+TB of images using 20,000+ lines of Python and C
- Implemented real-time neural network scoring of planet candidates in Dockerized Python Flask apps
- Engineered HPC scheduler to execute 1,000+ compute processes via many-to-many SQL databases
- Measured accuracy and completeness by applying statistical methods to artificially injected signals

Parallelized Executable for Removing Noise from Telescope Images
Lawrence Berkeley National Laboratory • September 2019 - June 2020 w/ Dr. Peter Nugent

- Constructed physical models of atmospheric fringes in optical images with principle component analysis
- Built parallelized feature identification and extraction tool currently running on 50,000+ images per night
- Released code as an open source Python package: *fringez* fringez Documentation

Education

PhD, Astrophysics, University of California: Berkeley May 2021
Advisors: Jessica Lu and Peter Nugent
Thesis: *Discovery of Rare Signals in Large Scale Time Domain Surveys: Dark Planets and Black Holes*

MA, Astrophysics, University of California: Berkeley 2017

BS, Physics and Astronomy, Northwestern University 2011
Weinberg College of Arts and Sciences
Advisor: Dr. Michael Smutko, Collaborator: Dr. Vicky Kalogera

BS, Theatre, Northwestern University 2011
School of Communication

Public Software Packages

PopSyCLE (2020) PopSyCLE Documentation
Population Synthesis for Compact-object Lensing Events

- Developed pipeline infrastructure to execute code in high performance supercomputing environments
- Reconfigured data format schema to extendible compound HDF5 to enabling additional image filters

zort (2019) zort Documentation
ZTF (Zwicky Transient Facility) Object Reader Tool Sole Author

- Executed spatial cross-matching and filtering for time-domain measurements of billions of objects
- Official data reader for ZTF telescope Public Data Releases, representing 11+ international institutions

Awards

UC-National Lab In-Resident Graduate Fellowship
The Regents of the University of California 04/2019 - 03/2021

- Awarded $130,000 to discover isolated black holes at the Lawrence Livermore National Laboratory

Interests

Astrophotography, Racquetball, Chess, Theatre, Financial Coaching